SOURODEEP KUNDU

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M.TECH CSE (Core), VIT Vellore | 2027 (ongoing) | CGPA 9.28/10
B.TECH CSE, KIIT DU | 2025 | CGPA 8.08/10



QS World Rank(CSIS): 110

Professional Summary

A motivated and research-driven technologist pursuing a **Master's in Computer Science Engineering (Core)**, specializing in the interdisciplinary convergence of **AI, Quantum Computing & LLMs**. Skilled in **C/C++ and Python** algorithm development across **ML, DL, GenAI, QML and Agentic AI**. Proficient in AI model engineering, evaluation, data-driven automation and Android/full-stack healthcare development, with a background in Quantum Circuit Optimization, QKD & Quantum Information Processing. Builds efficient, well-documented software through modular design and collaborative development. Computational & experimental research spans clinical AI, quantum simulations and hybrid ML; current work explores **LLM distillation and quantization with Ollama/vLLM**.

Education

VIT, Vellore **M.Tech. CSE (Core, ongoing)** | CGPA: 9.28/10, **Transcripts** *2025 – 2027*
Department Rank 3; Chancellor's Merit List Award. Coursework: Algorithms, Architecture, Operating Systems, ML, Networks, DevOps.
KIIT, DU Bhubaneswar **B.Tech. CSE** | CGPA: 8.08/10, **Transcripts** *2021 – 2025*
Mumbai: Narayana Junior College – **HSC XII**: 93.67% (2021) | D.A.V Public School – **CBSE X**: 92.8% (2019)

Technical Skills

Programming & Systems: C/C++, Python, Java, JavaScript/TypeScript, SQL; OOP, DSA, API design, Git/CI/CD, AWS/Azure.
Parallel Computing: CUDA, OpenMP, CUDA-Q – foundations and personal practice projects.
Android & Full Stack: Bluetooth, JNI/AIDL, lifecycle management, SQLite, ADB, Gradle/Maven; **React/Next.js, FastAPI**, Node.js, REST APIs, async services, SSE, JUnit/JaCoCo.
Agentic AI & LLMOps: LangChain/LangGraph, CrewAI, MCP, ReAct/tool calling, context engineering, **hybrid RAG**, RRF/FTS5, graph retrieval, retrieval evaluation; LlamaIndex, Ollama, Hugging Face; vLLM/LiteLLM serving design.
ML & Clinical AI: PyTorch, TensorFlow/Keras, scikit-learn, JAX, OpenCV, Pandas/SciPy; multimodal and federated learning, **patient-disjoint validation**, ablations, uncertainty estimation and explainability.
Quantum & Medical Software: Qiskit, PennyLane, Cirq, SeQUeNCe; VQC/QNN, kernels, QFIM, QUBO/QAOA, annealing, tensor networks, circuit mapping, hardware/noise studies; **IEC 62304** traceability.

Research & Engineering Experience

SDE Apprentice – AI, Healthcare & Android | ***PHILIPS R&D, AM&D, Bengaluru*** *May 2026 – Present*
- Develop **Android cardiac telemetry (MCOT)** software connecting wearable ECG sensors with patient mobile monitors.
- Build **Bluetooth and Java/native integration** through JNI/AIDL interfaces, modular services and lifecycle management; simplify device setup while preserving existing hardware workflows.
- Implement **multi-device diagnostic context**; prototype medication prompts and warning controls; design patient-status/recovery views and investigate ECG data recovery.
- Automate **IEC 62304 specification traceability** and revision review; strengthen JNI integration testing and CI.

AI Platforms & Agentic SDLC | ***Philips R&D / AM&D*** *2026*
- **C6 Brain:** built **engineering RAG and an agentic SDLC harness** over code, requirements and QMS; integrated source citations, hybrid search and FastAPI/MCP with scoped tools and approval gates.
- **VajraDocAI:** built **full-stack document automation** to reduce QMS/SOP review effort: field extraction, revision tracking, source-linked change plans and Word review copies with approval workflows.
- **PolyMind:** built multi-agent orchestration with streaming, model routing and Ollama/GPU inference; integrated **CrewAI/LangGraph** checkpoints, approval interrupts and execution traces for **DocGen/App Builder**.
- **RAG evaluation:** built hybrid keyword/sparse-vector retrieval with **RRF** and a harness to measure ranking, citation grounding, abstention and latency: precision/recall@k, MRR/nDCG and p95 latency.
- **Migration Intelligence:** automated work-item and Wiki mapping, with traceable ticket-to-code workflows.
- **Org AI – Philips LLM initiative:** built governed LLM app-generation and expense-tool workflows with checkpointed plans, staged changes, verification and recovery through CLI and graphical interfaces.
- **Model Bridge:** built a Copilot/Ollama MCP broker with model discovery, validation and isolated adapters.
- **Mobile AI assistant:** prototyped SLM chat with RAG over product guidance for support.

Selected Research | ***VIT Vellore / Research Collaborations*** *2025 – Present*
- **QVoC:** co-developed utility-gated leukemia inference with patient-disjoint, cross-fitted evaluation against classical/neural early-exit baselines; assessed **IBM Quantum circuit execution**.
- **HQFL-QNet:** co-developed compressed-feature federated learning with QUBO selection; evaluated heterogeneous-client classical controls, with separate **IBM Quantum circuit/noise studies**.
- **AURA-Q:** built a governed quantum-processor selector with typed intent, feasibility checks and abstention; evaluated resource-aware routing in noisy simulation.

Research & Professional Experience (continued)

AI Models & Quantum Systems | ***Research Projects & Collaborations*** *2025 – Present*
- **Parallel-computing practice:** implemented OpenMP scheduling, reductions and synchronization (**Demo**); developing a C++/OpenMP look-ahead circuit router.
- **Speech/QML prototype:** built a PyTorch CNN–PennyLane attention pipeline on synthetic 10-class log-Mel inputs; explored speech-command modeling through correspondence with **Dr. Vinayak Abrol, IIIT Delhi**.
- **QL-GASNet:** investigated generative augmentation and dual quantum kernels for leukemia classification.
- **QC-GENScan:** developed CT-patch encoding and variational-circuit/Swap-Test methods for brain-tumor screening.
- **NeuroQ-Wound:** proposed segmentation/biomarker fusion and calibrated, interpretable risk scores for clinical review.
- **RetinaQ-SAGE / RetiQ-Trace:** prototyped quantum/classical retinal-response forecasting; evaluated patient-disjoint splits, reset ablations and bootstrap intervals.
- **EQuR-Mol:** simulated selective molecular inference with scaffold splits, classical controls and quantum-call budgets.
- **QRS-Capsule:** designed reversible image custody with original recovery, consent policies and post-quantum key protection.
- **DEQUINT:** simulated entanglement generation, teleportation and wireless quantum interconnects on a 4×4 core mesh; analyzed network topology, latency, fidelity and throughput.
- Co-authored a **multimodal tumor survey** on radiomics, deep-learning fusion and quantum classifiers for tumor assessment.
- **Q-SDLC/ISEC:** proposed a quantum-software development process. **QML-finance:** designed portfolio/risk studies with walk-forward evaluation and classical baselines.

Quantum Cloud & Edge Computing Researcher (Internship) | ***C-DAC INDIA**, Chennai* *Sep 2025 – May 2026*
- Developed **QML-AURA** for quantum/cloud-edge scheduling using **CUDA-Q**, QUBO, QAOA/annealing adapters, fairness constraints and classical fallback.

T&T Cyber-Strategy: D&R Threat Analyst (*Offer*) | ***DELOITTE**, Mumbai* *Dec 2025*
- Undertook Deloitte D&R onboarding and CySA+ training; received PPO and deferred further engagement for M.Tech.

Project Assistant | ***Kalinga Institute of Industrial Technology*** *Dec 2022 – Aug 2025*
- **Major thesis / QLSTM quantum finance:** developed hybrid neural models with variational quantum layers for stock forecasting under Dr. Nachiketa Tarasia; published in IEEE Xplore.
- **Minor thesis / cryptanalysis:** studied GA/PSO for transposition-cipher decryption and n-gram ciphertext-only analysis under Dr. Partha Pratim Sarangi.
- **Quantum circuits:** with Dr. Anirban Bhattacharjee, collaborated over two years across 15+ projects on nearest-neighbor synthesis, qubit ordering, routing and SWAP-cost optimization.
- **Laryngeal-cancer AI:** investigated image/voice/NLP fusion for early detection with Dr. Debajyoty Banik.
- **Cloud systems:** analyzed load-balancing algorithms and distributed-workload performance with Dr. Ronali Padhy.
- **AI-SCOPE:** collaborated on ML, cloud and cryptography during incubation with Dr. Mrutyunjay Suar / KIIT-TBI.

CS Associate (*Link*) | ***Recruit CRM, New Jersey, USA*** *Feb 2025 – Jul 2025*
- Automated onboarding/chatbots via Workato/Zapier and Intercom APIs; supported HNA, LATAM and Europe.

Research Internship (*Link*) | ***IIT BOMBAY, LOQM*** *Nov 2024 – Feb 2025*
- Researched programmable photonics and waveguides under **Dr. Anshuman Kumar**; proposed **Optics-Informed Neural Networks** combining Maxwell-informed learning, quantum materials and neuromorphic optics.

Co-founder & CTO (*Link*) | ***AI-SCOPE, KIIT-TBI*** *Sep 2024 – Jan 2025*
- Built **Adaptive Intelligence for Sports-biomechanics Correction & Performance Enhancement**; AI injury-prevention analytics with Dr. Nachiketa Tarasia (EAM) / Dr. Sudeep Sathpathy (KIMS). Sports-pressure device IP/design **435501-001**.

Cyber Threat Intelligence Analyst Intern (*Link*) | ***I4C, Ministry of Home Affairs, Govt. of India*** *2025*
- Worked on VAPT, network security, OSINT, deepfake detection and blockchain forensics; developed LangChain-based cyber-investigation workflows at **I4C** under **Mayank G., Deputy Director**.

Quantum Computing Intern (*Link*) | ***Creed&Bear, UAE*** *Aug 2024 – Oct 2024*
- Developed quantum models for portfolio and stock forecasting under **Flavio Villa**.
- Designed annealing-based feature selection using **D-Wave Ocean**.

Research Intern (*Link*) | ***BITS PILANI, CSIS*** *Aug 2024 – Sep 2024*
- Researched **Quantum Ray Marching** for light-transport simulation.

Major Project Intern (*Link*) | ***HIT KOLKATA, CSE*** *Aug 2024*
- Studied evolutionary nearest-neighbor circuit mapping and **SWAP reduction** with **Dr. Anirban Bhattacharjee**.

S.N. Bose Intern (*Link*) | ***NIT SILCHAR*** *Jun 2024 – Jul 2024*
- Explored **MPS/PEPS/MERA tensor networks** to represent quantum states and study hybrid-QML circuit optimization; used Qiskit/PennyLane under **Dr. Anish Kumar Saha**.

Summer Research Intern (*Link*) | ***IIT GUWAHATI***, **Paper** *May 2024 – Jul 2024*
- **MARS Lab:** simulated multicore quantum interconnects, wireless NoC, qubit allocation and entanglement/teleportation in *SeQUeNCe* under Dr. John Jose; presented findings to Dr. Prabhat Mishra.

QML Research Internship (*Link*) | ***IISER BHOPAL*** *2024*
- Explored generative QML for high-dimensional data and quantum error correction under **Dr. Kuntal Roy**.

ANRF (formerly SERB) Project Associate (*Link*) | ***IIT PATNA*** *Dec 2023 – Jan 2024*
- Contributed to **SERB**-supported quantum-neuromorphic/QKD research on secure communications and encryption.

Summer Research Intern (*Link*) | ***IIEST SHIBPUR*** *May 2023 – Jun 2023*
- Under **Dr. Hafizur Rahaman**, implemented C++/Python **1D/2D circuit transformation**, graph/genetic qubit ordering, **SWAP insertion and look-ahead mapping**; simulated and validated transformed layouts.

Publications – Conference Proceedings & Book Chapters

S. Kumar, **S. Kundu**, H. Rahaman, A. Bhattacharjee. *Enhanced Quantum Circuit Layout Optimization: A 2D Nearest Neighbour Approach Leveraging K-Means Clustering and A* Swapping*.
Springer **COMSYS 2024** (2026), BITS Pilani, Goa Campus. pp. 531–547. **PDF** · **DOI**

S. Kundu, D. Mishra, N. Tarasia, R. K. Lenka, R. K. Barik. *QEA-FAD: Quantum Enhanced Autoencoders for Anomaly Detection in Finance and Healthcare*. IEEE **NETCRYPT 2025**, JNU, New Delhi. pp. 1280–1285. **PDF** · **DOI**

S. Kumar, **S. Kundu**, H. Rahaman, A. Bhattacharjee. *A 2D Transformation Technique for Nearest Neighbor Realization of Quantum Circuits*. Springer **VDAT 2023** (online 2024 / print 2025), BITS Pilani / CSIR-CEERI, Pilani. pp. 273–287. **PDF** · **DOI**

S. Kundu, D. Mishra, N. Tarasia, R. K. Barik. *QEFMLA-FIDO: Quantum Evolution Formulation for ML Advancement: Fidelity-Improved Data Optimization in Quantum-Classical Systems*. IEEE **AESPC 2024**, KIIT, Bhubaneswar. pp. 1–6. **PDF** · **DOI**

S. Kundu, N. Tarasia, R. K. Barik. *QLSTM4FM: Quantum Assisted Long Short-Term Memory Framework for Financial Market Trend Forecasting*. IEEE **IC-SIT 2024**; **Best Paper**, Silicon University, Bhubaneswar. pp. 1–6. **PDF** · **DOI**

S. Kundu, S. Kumar, L. Biswal, C. Bandyopadhyay, A. Bhattacharjee, H. Rahaman. *An Improved Circuit Transformation Technique for Nearest Neighbor Implementation of Quantum Circuits*. IEEE **VDAT 2024**, VIT Vellore. pp. 1–6. **PDF** · **DOI**

S. Kundu, S. Kumar, H. Rahaman, A. Bhattacharjee. *A 2D-Based Synthesis Strategy for Nearest Neighbor Transformation of Quantum Circuits*. Springer **COMSYS 2023** (2024), IIT Mandi. pp. 591–614. **PDF** · **DOI**

S. Kundu, A. Bandyopadhyay, A. Bhattacharjee. *Mapping of Nearest Neighbor-Based Quantum Circuits into 2D*.
CRC Press, Big Data Computing (2023). pp. 249–265. **PDF** · **DOI**

Journal Manuscripts & Preprints

Manuscripts | ***Quantum Computing / Clinical AI*** *2026*

- **QVoC:** Learning the Value of Quantum Computation for Selective Hybrid Inference.
- **HQFL-QNet:** Hybrid Quantum Federated Learning with QCNN-AAE Representations and Tensor-Based Optimization.
- **DEQUINT manuscript:** A Decentralised Quantum Interconnect with Bidirectional Teleportation and Wireless NoC for Scalable Multi-Core Quantum Architectures.
- **QML-AURA:** Adaptive Unified Resource Allocation with Hybrid QUBO–QAOA Optimisation and Multi-Agent Orchestration.
- **Co-authored survey:** From Radiomics to Quantum Circuits: A Critical Survey of Multi-Modal Fusion Architectures and Algorithms Across Classical, Deep, and Quantum Intelligence for Tumor Diagnosis.

QC-GENScan | ***Quantum Circuitry for Genetic Encoding Network and Imaging-Based Brain Tumor Screening*** *2025 – 2026*
(*ICSDP 2025* · *PDF* · *Best Paper*)

Preprint | ***Q-PACT: Quantum-Parallel Agentic Coordination Toolkit*** *Jan 2026*

- Sole-authored Research Square preprint on fairness-aware quantum/agentic scheduling (Jan 2026). **DOI**

Patents & Intellectual Property

Published Patent Applications | ***Quantum Computing / AI Inventions*** *2026*

- **Selective quantum assistance for edge-device remaining life: 202641102793** (26 Aug 2026), with Jayashree.
- **Secure quantum federated learning: 202641096866** (11 Aug 2026), with Kalyanaraman P and Lokeshkumar Ramasamy.
- **DEQUINT: 202641068186** — decentralized quantum interconnects and teleportation.
- **Published project applications: QML-AURA, QL-GASNet, NeuroQ-Wound, RetinaQ-SAGE, QRS-Capsule, QC-GENScan**.

DESIGN | ***AI-BASED DEVICE FOR DETECTION OF SPORTS PRESSURE INJURIES*** *2024*

- Designed an AI-SCOPE device for adaptive sports-pressure-injury analysis; Indian design **435501-001**.

Honors & Professional Development

Samsung Scholar (Grade II), IISc/Synopsys/Samsung ISWDP: Levels 1–2 with distinction (**92%**, 2026). Programmes: IISc quantum, Kotak-IISc AI/ML Fintech (2026) and QWorld (2025).

AI curriculum: 12 modules on agents, MCP, FastAPI, vLLM/LiteLLM, OWASP and OpenTelemetry/Prometheus.

QLSTM4FM, IC-SIT 2024 **Best Paper**.

Certificates & workshops: *Link*; *ISRO*; *PPT*; *Link*.

Research Offers & Selections

IIT Kharagpur (2026): Smart ECG Holter research selection with **Dr. Debanjan Das** (Affordable Healthcare).

IIIT Delhi research internship offer (2026): speech/audio QML with **Dr. Vinayak Abrol, Cross-CapsLab**.

ACM India iSOFT Ascent (2026): selected; assigned mentor **Prof. Y. Raghu Reddy, IIIT Hyderabad**, programme coordinator.

BITS–RMIT PhD selection (2024): quantum-resilient space cybersecurity; Dr. Ashutosh Bhatia / Dr. Abebe Diro.

IIT Ropar research internship (2024): neuromorphic, drug-discovery and dental ML; Dr. Sukrit Gupta.

NIT Rourkela SIP (2024): Grover-based Max-Cut optimization; Dr. Shyamapada Mukherjee.

Research interests: LLM-as-a-Judge, reliable multilingual NLP, privacy-preserving analytics and resource-aware quantum AI.